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Course: Calculus Beta

Assignment: MyLab Math - Uge 7

Use the union rule to answer the question.

If $n(B) = 52$, $n(A \cap B) = 15$ and $n(A \cup B) = 86$, what is $n(A)$?

Recall that the union rule for sets states that

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

for any finite sets A and B . To find $n(A)$, substitute each of the given quantities into the union rule. Then, solve for $n(A)$.

Substitute the given values of $n(B)$, $n(A \cap B)$, and $n(A \cup B)$ into the union rule.

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ 86 &= n(A) + 52 - 15 \end{aligned}$$

Now, solve for $n(A)$.

$$\begin{aligned} 86 &= n(A) + 52 - 15 \\ 49 &= n(A) \end{aligned}$$